HARI PRASAD GUNTUKU

AI / ML Engineer | NLP | Embeddings | Time-Series Forecasting | Python

Hyderabad, India | +91 8555081508 | connect.hpg@gmail.com

linkedin.com/in/dot-hpg | github.com/dot-hpg | Portfolio: hariprasad-ai-engineer.vercel.app

PROFESSIONAL SUMMARY

AI/ML engineer with 2+ years of industry experience as a Data Analyst (SQL and Python pipelines, data validation, Power BI), now building end-to-end ML systems in NLP, embeddings and forecasting. Takes models from notebook to tested, deployed product: fine-tuned BERT for tweet NER (3.2x a BiLSTM baseline), a LightGBM forecaster for 365 stores (16.8% daily WAPE), and an on-device embedding app with ~97% test coverage. Pursuing an M.S. in Computer Science (AI & ML).

TECHNICAL SKILLS

ML & NLP: BERT fine-tuning, BiLSTM-CRF, Sentence Embeddings, TF-IDF, NLTK, Scikit-learn, TensorFlow, LightGBM, XGBoost, SHAP
Modeling & Statistics: Time-Series Forecasting, Rolling-Origin Validation, Regression, Hypothesis Testing, Bootstrap, ROC AUC
Deployment & MLOps: FastAPI, Flask, Streamlit, Docker, Kubernetes, ONNX Runtime Web, Transformers.js, GitHub Actions CI/CD
Languages & Data: Python, SQL (PostgreSQL, MySQL, BigQuery), TypeScript, React, Pandas, NumPy, Power BI, Tableau

PROFESSIONAL EXPERIENCE

Data Analyst | Ivue Technologies Pvt. Ltd., Hyderabad — Mar 2024 – Present

- Built and automated SQL and Python data pipelines (joins, CTEs, window functions) for extraction, transformation and reporting across multiple business units, with validation checks keeping reports consistent with source data.
- Automated recurring reporting and developed Power BI dashboards and KPI reports, working with stakeholders to define requirements, metric logic and refresh cadence.
- Analyzed large operational and customer datasets with Python and Excel; built RFM segmentation and customer lifetime value (CLV) models to identify high-value customer groups.

AI / ML PROJECTS

FitCheck: On-Device AI Resume Matcher | *Transformers.js, ONNX, React, TypeScript* — github.com/dot-hpg/fitcheck

- Shipped a private resume-to-job matcher running the all-MiniLM-L6-v2 embedding model (~23 MB quantized ONNX) fully in the browser via a Web Worker, with an explainable 100-point score and skill-gap analysis over 350+ skill aliases.
- Reached ~97% unit-test coverage of the matching engine plus Playwright end-to-end tests (including offline real-model runs); GitHub Actions lints, type-checks, tests, builds and deploys on every push.

Twitter Named Entity Recognition | *BERT, TensorFlow, FastAPI, Docker, Kubernetes* — github.com/dot-hpg/twitter-ner

- Ran 17 controlled experiments on WNUT-16 comparing BiLSTM-CRF with fine-tuned BERT for entity extraction from noisy tweets; fine-tuned BERT reached test F1 0.437, 3.2x the LSTM baseline.
- Served the best model as a containerized FastAPI service and published a technical write-up on Medium.

Retail Sales Forecasting | *LightGBM, XGBoost, SHAP, Flask, Streamlit* — github.com/dot-hpg/sales-forecasting

- Forecast daily sales 61 days ahead for 365 stores; compared 10 models and tested 8 hypotheses using leakage-free features and rolling-origin validation.
- Global LightGBM hit 16.8% daily WAPE on a 61-day hold-out (~90% accurate at store-month level) and measured a +33% discount sales lift; explained with SHAP, served via Flask API and Streamlit.

FlipItNews: News Classification | *Scikit-learn, NLTK, TF-IDF, FastAPI* — github.com/dot-hpg/flipitnews-news-classification

- Compared 12 model and vectorizer pairs with cross-validation and leakage checks; Logistic Regression reached 98.0% test accuracy, rising to 99.8% by routing low-confidence articles to an editor; served via FastAPI in Docker.

DATA SCIENCE CASE STUDIES

- **Target Brazil (SQL, BigQuery):** 99,441 orders; +137% YoY order value; very late deliveries cut reviews from 4.32 to 1.70.
- **LoanTap Credit Risk (Logistic Regression, SMOTE):** 396,030 loans, 0.70 ROC AUC; 81% accuracy masked just 7% defaulter recall.
- **Walmart Black Friday (CLT, Bootstrap):** 550,068 transactions; traced the ~8% gender spending gap to product mix.

8 more case studies (Yulu, Delhivery, Netflix, Aerofit, Jamboree and others) on github.com/dot-hpg

EDUCATION

M.S. in Computer Science (AI & ML) | Woolf University, via Scaler Neovarsity — In Progress
Advanced Data Science & Machine Learning Program | Scaler Academy — In Progress
Bachelor of Architecture (B.Arch), First Class | School of Planning and Architecture, New Delhi

Certifications: HackerRank SQL (Intermediate, Basics) | NSDC: Python Libraries, SQL, EDA Fundamentals